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Installation Instructions

Model ZAM-300 (50V)

Zone Amplifier Module 300W (S54505-B17-A1/A2)

INTRODUCTION

The **Siemens** Model ZAM-300 is a Zone Amplifier Module with up to 300 Watt constant output power. The output voltage can be set (connector) to 50 VRMS.. The ZAM-300 provides two separate speaker line circuits. The two speaker lines can be configured (Zeus) as one Class A zone (loop) or one Class A/B zones (two separate supervised and protected speaker lines). The Class A/B configuration is typically used to drive two speaker lines in-between one speaker zone.

The ZAM-300 can be configured (Zeus) as a backup amplifier. In Zeus a 1 to 1 or a 1 to many (up to 24 main) backup can be selected. One ZAM-300 backup amplifier can backup one ZAM-300 main at a time or up to four ZAC-70 main at a time.

The ZAM-300 provides a local analog (0dB) audio input circuit and an input to activate the amplifier via a dry contact.

The ZAM-300 provides an analog insert to loop-in E100 specific audio processing equipment (e.g. APC-4).

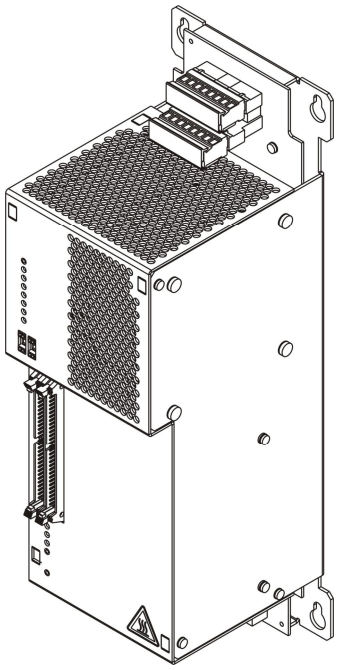


Figure 1 ZAM-300

The ZAM-300 is connected to the control bus (CAN), 8 channel digital audio bus (ASI) and degrade alarm via the 60 pin backplane bus.

The ZAM-300 mounts at the back of the enclosure of the system and occupies ¼ of the total space of the Mounting Plate.

Features

ZAM-300 features are as follows:

- Audio amplifier with 300 Watt constant output power in Class D technology
- Over all efficiency at least 85% between ¼ to full load
- Amplifier is protected against open, short circuit, over current, over heating and over modulation
- Amplifier is supervised for functionality
- Both speaker lines are separately supervised for open and short circuit
- Both speaker lines are supervised for ground fault
- The ZAM-300 is driven by a micro controller
- The digital audio bus (ASI) is driven by a FPGA
- Software update via Zeus/PMI

OPERATION

The ZAM-300 receives its control (CAN) from the PMI via DAC-NET.
ZAM-300 sends the status data via DAC-NET to the PMI.
One ZAM-300 takes one sub-address of the DAC-NET.

The transmission of audio data from the DAC-NET to the ZAM-300 is realized via a proprietary digital audio bus named ASI (Audio Serial Interface).
Transmission on the ASI is executed with a timeslot method and is synchronized with Frame and Bit Synchronize Signals. Via control data on the CAN Bus, the timeslots are allocated to each ZAM-300. Each ZAM-300 is configured to listen to its corresponding timeslot.

A local analog audio input (0dB) is provided to connect an external audio source. The audio input is synchronous (isolated via transformer) to prevent ground loops. The local audio input can be activated via an external dry contact.

The analog insert provides the ability to loop in E100 specific audio processing equipment, such as equalizer, delay and automatic sound pressure level adjustment.

The two speaker lines are individually supervised for open and short circuit. The two speaker lines are individually switched off in case of short.
Ground fault detection circuit is only one time on board for both speaker lines – only one trouble message for both lines.
Speaker lines are supervised in de-active and active state.

Each speaker on the line must be isolated for dc current via a bipolar capacitor.

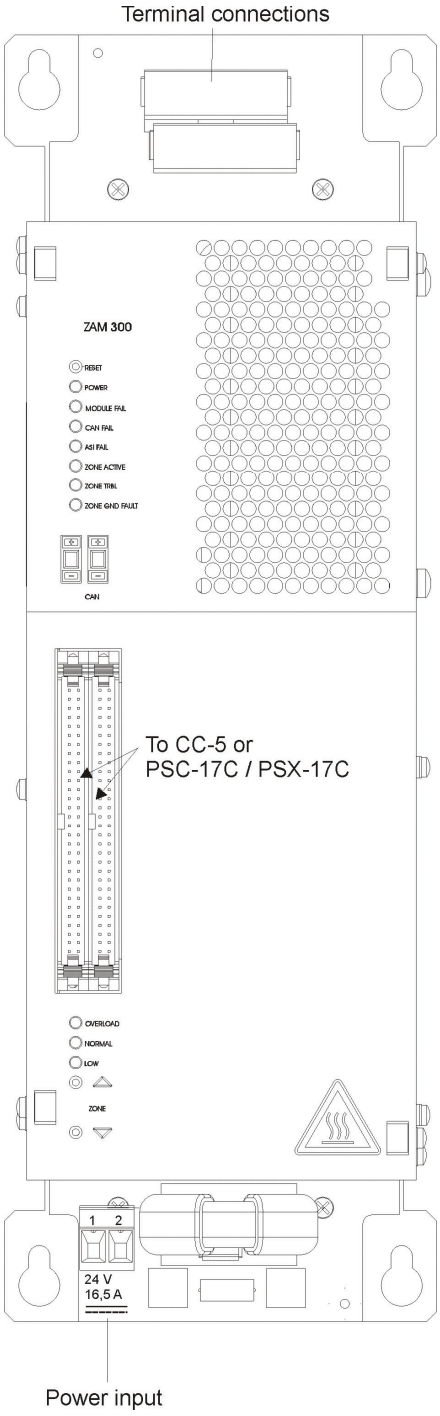


Figure 2
Terminal Blocks, Controls and Indicators

Controls and Indicators at the ZAM-300 front panel (see ***FFigure 2 Terminal Blocks, Controls and Indicators***.)

The ZAM-300 front panel provides one Reset switch, seven diagnostic LEDs, two CAN address switches, two push buttons for the local audio input level adaptation and two 60 pin connections.
The top of the module provides two terminal connections.
The button of the module provides the power input block.

Reset Switch

A reset switch is located on the top of the front panel. Pushing the reset switch re-initializes the ZAM-300 operation. During re-initialization the individual configuration (dependant on the CAN address) is automatically download from the DAC-NET

Diagnostic LEDs and level adaption unit LEDs are defined as follows:

LED	COLOR	NORMAL STATE	OFF NORMAL STATE	ACTION IN OFF NORMAL STATE
POWER	green	ON	OFF –external voltage fail	Check for failure on PSC-17C, Power input wiring
MODULE FAIL	yellow	OFF	ON –general trouble at ZAM-300	Failure on ZAM-300, ZAM-300 needs to be replaced
CAN FAIL	yellow	OFF	ON –CAN interface trouble	Failure on AIC, DAC-NET, CC-5 backplane, 60 pin flat ribbon cable or other CAN cards/modules
ASI FAIL	yellow	OFF	ON –ASI interface trouble	Failure on “ASI OUT“ and/or “ASI IN“ at this ZAM-300, CC-5 backplane, 60 pin flat ribbon cable or other ASI cards/modules
ZONE ACTIVE	green	OFF	ON –Speaker zone active	Indicates normal operation on speaker zone
ZONE TRBL	yellow	OFF	ON –Speaker zone trouble, open circuit or short circuit	Check wires to the speakers, EOL resistor(s) and de-coupling capacitors on every speaker
ZONE GND FAULT	yellow	OFF	ON –unwanted earth ground connection on a speaker line	Check wires to the speakers for earth ground connections
OVERLOAD	yellow	OFF	ON	Refers to Local Audio input only: Press “DOWN” push button until “OVERLOAD” LED turns off and “NORMAL” LED turns on
NORMAL	green	ON	OFF	Refers to Local Audio input only: No audio signal or audio signal too high/low
LOW	yellow	OFF	ON	Refers to Local Audio input only: Press “UP” push button until “LOW” LED turns off and “NORMAL” LED turns on (see section Volume setting for Local Audio In)

CAN Address Switch

Set the CAN address according to the setting in the configuration tool Zeus.

Volume setting for Local Audio In

Use the ZONE buttons ▲ (up) and ▼ (down) to adopt the audio volume of the external device.

How to find the optimal setting:
Press the ▲ button until the yellow OVERLOAD LED lights up.
Now press the ▼ button until the OVERLOAD LED is off and the green NORMAL LED lights up.

A single press of the ▲ (up) or ▼ (down) button will change the volume of about 1dB. Holding down the buttons will constantly change the volume in steps of 1dB until the end of the setting range is reached.

Attention:
Reaching the end of the level setting range is not visible
Value of the audio setting is not visible
Value of the audio setting is only stored in the EEPROM of the amplifier itself, not in the PMI database. After a replacement of the amplifier the volume setting of this local audio input is not automatically reloaded via the DAC_NET module. It has to be set again manually. If possible leave the setting at the default position (0 dB). Instead set the volume at the external device.

Terminal Connectors at the top of the module

These terminals are used for field wiring as shown in ***Figure 19 ZAM-300 Back-Up Amplifier wiring***

60 Pin Connectors

This connector connects the amplifier to the E100 control and digital audio bus.

Power Input Terminal Connector at the bottom of the module

This 24V power input terminal of the ZAM-300 is connected to the 24 V/17 A power output terminal TB4 of the PSC-17C/PSX-17C as shown in ***Figure 14 ZAM-300 24V Wiring***.

The ZAM-300 takes one sub-address of the DAC-NET. To configure the CAN address use the two rotary-switches on the front panel (refer to ***FFigure 2 Terminal Blocks, Controls and Indicators***)

In order to change the jumper settings remove the housing first.

NOTE

If the amplifier is continually operated with tones at full power the housing may become hot.

Remove the 6 screws on the left and the right side of the ZAM-300 housing as shown in ***Figure 3 Position of the 6 Screws which hold the cover*** to remove the cover.

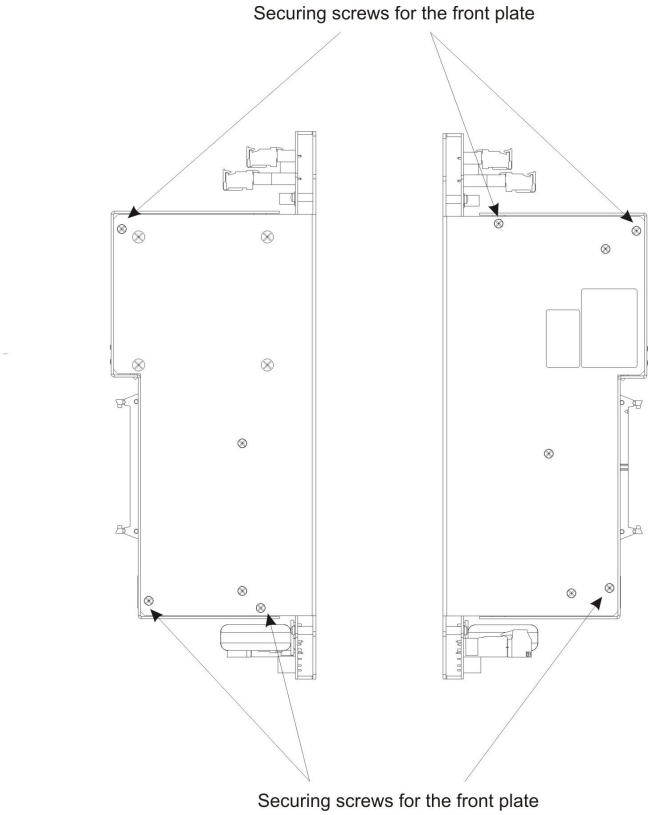


Figure 3 Position of the 6 Screws which hold the cover

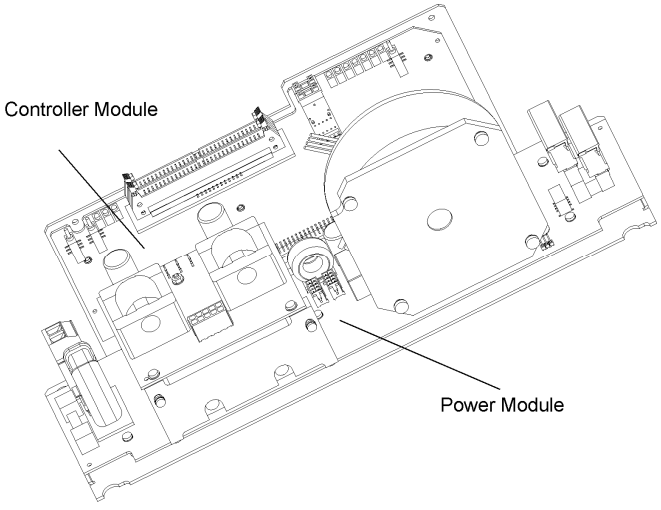


Figure 4 ZAM-300 Module inside

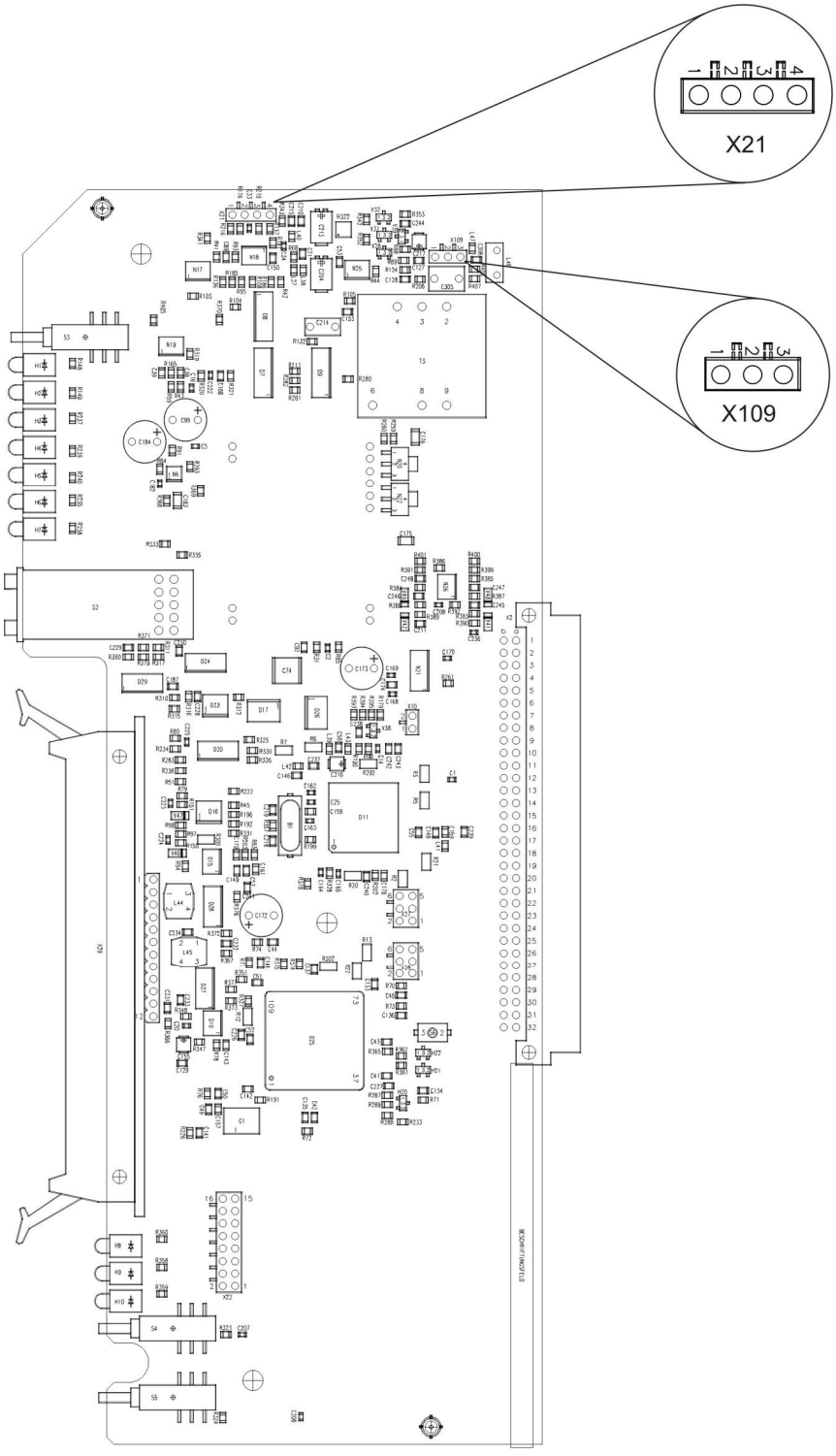


Figure 5 Position of the pins to set the Low Frequency Cut off and the Insert Function

Setting of the Low Frequency Cut Off

The low frequency cut off is set default to 130 Hz. It can be set to 240 Hz. The cut off setting is true for both audio sources: internal audio channels and local audio input.

Cut off at 240 Hz protects low quality speakers from being overloaded by low frequency pulses with high power. It can reduce the amplifier power and increase the speech intelligibility.

For music reproduction with higher quality leave the low frequency cut off at default (130 Hz).

130 Hz: no jumpers set at socket board X21

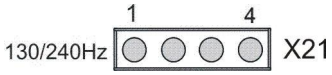


Figure 6 Jumper Position 130 Hz

240 Hz: two jumpers have to be set at socket board X21:
- between pin 1 and 2
- between pin 3 and 4

NOTE

Never set only one jumper.

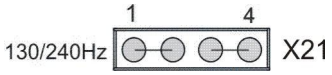


Figure 7 Jumper Position 240 Hz

Setting the Audio Insert Function

Insert Off: one jumpers has to be set at socket board X109
- between pin 1 and 2

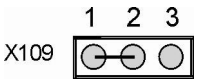


Figure 8 Jumper Position Insert Off

Insert On: one jumpers has to be set at socket board X109
- between pin 2 and 3

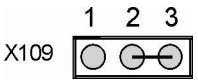


Figure 9 Jumper Position Insert On

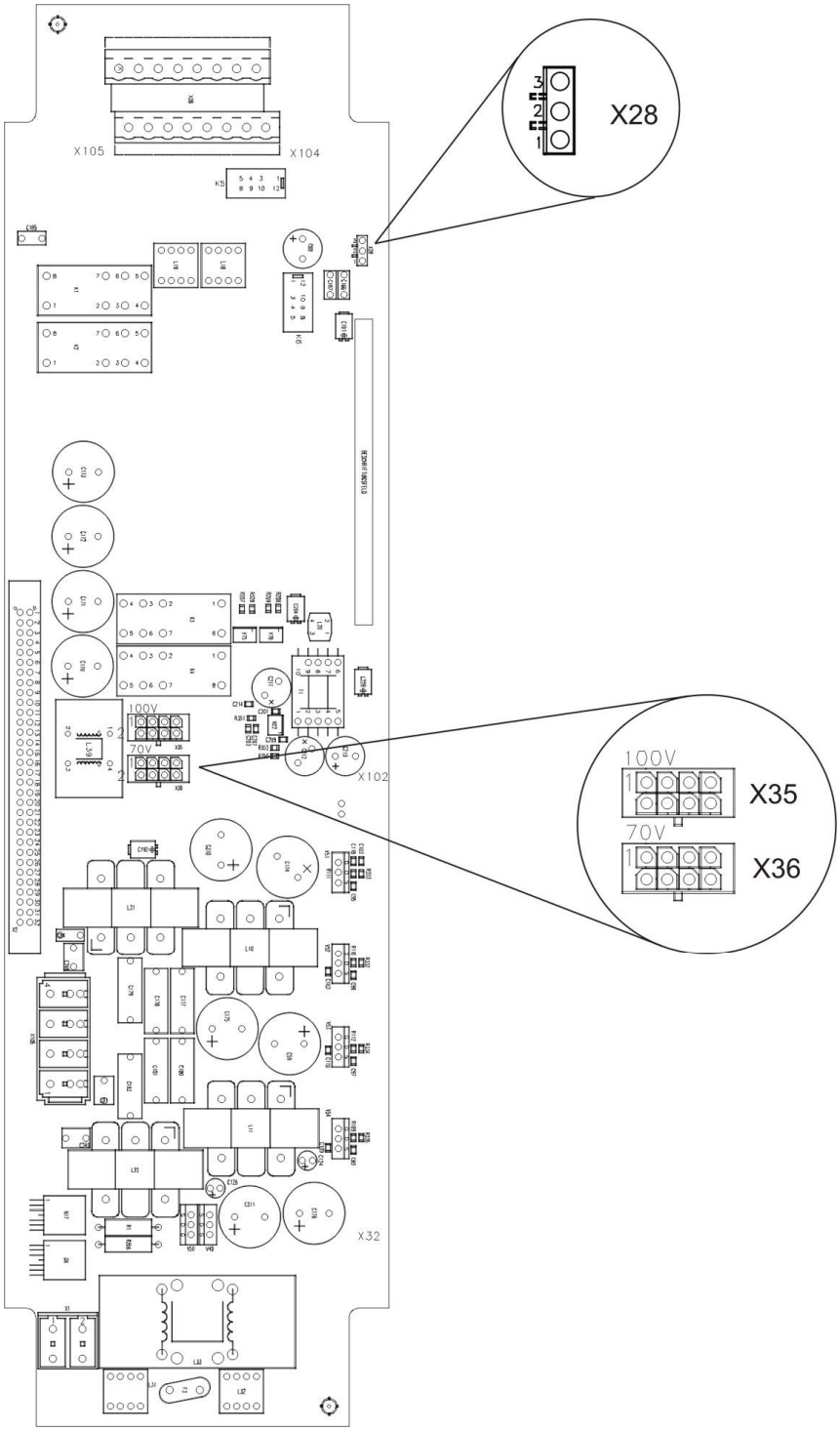


Figure 10 Position of Connectors and Jumpers to set the Speaker Line Voltage

The speaker line voltage is set at terminals X35 and X36
The amplifier supervision has to be set in according to the line voltage at the socket board X28.

Speaker Line Voltage 50V

Plug the connector from the output transformer to terminal X35 marked **100V** and set no jumper at socket board X28.

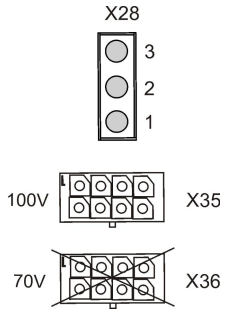


Figure 11 Jumper Position 100V

Mount the cover of the ZAM-300 and fix it with the 6 screws.

WARNING

Remove ELECTRICAL POWER prior to working on equipment.

NOTE

Take care of module temperature!

The ZAM-300 can occupy any position on the mounting plate. When the ZAM-300 is mounted correctly on the Mounting Plate it will be flush on the top, bottom, and right with the Mounting Plate and the mounting screw holes of the ZAM-300 will align with the threaded posts.

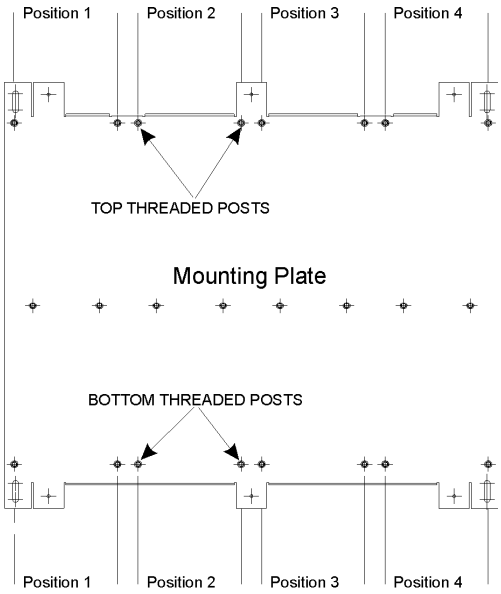


Figure 12 Location of the ZAM-300 at the Mounting Plate

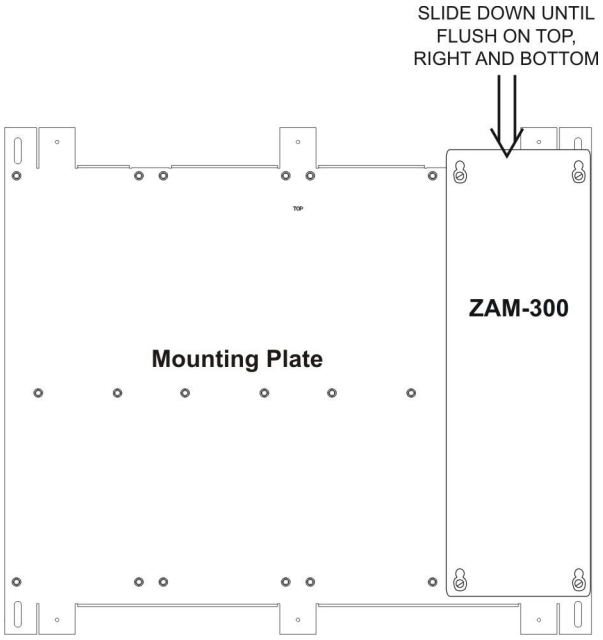


Figure 13 Mounting the ZAM-300 on the Mounting Plate

1. Install four M5 screws in the threaded posts at the mounting position. Screw each of the M5 screws into the threaded posts 5-6 turns.
2. Place the ZAM-300 over the four screws on the Mounting Plate and slide it down or towards you to rest on the four screws. When the ZAM-300 is in the correct position it will be flush with the top, bottom and right side of the Mounting Plate.
3. Tighten the four screws.

WARNING

Remove ELECTRICAL POWER prior to working on equipment.

All field wiring of the ZAM-300 is connected to the two terminal blocks on the upper end of the ZAM-300.

The screw terminals on the top can accommodate one 12-24 AWG (Ø 0.5 mm-2.5 mm²) or two 16-24 AWG, (Ø 0.5 mm-1.0 mm²).

The power input terminals at the bottom can accommodate one 10-24 AWG (Ø 0.5 mm-4 mm²) or two 12-24 AWG, (Ø 0.5 mm-2.5 mm²).

To connect external wiring

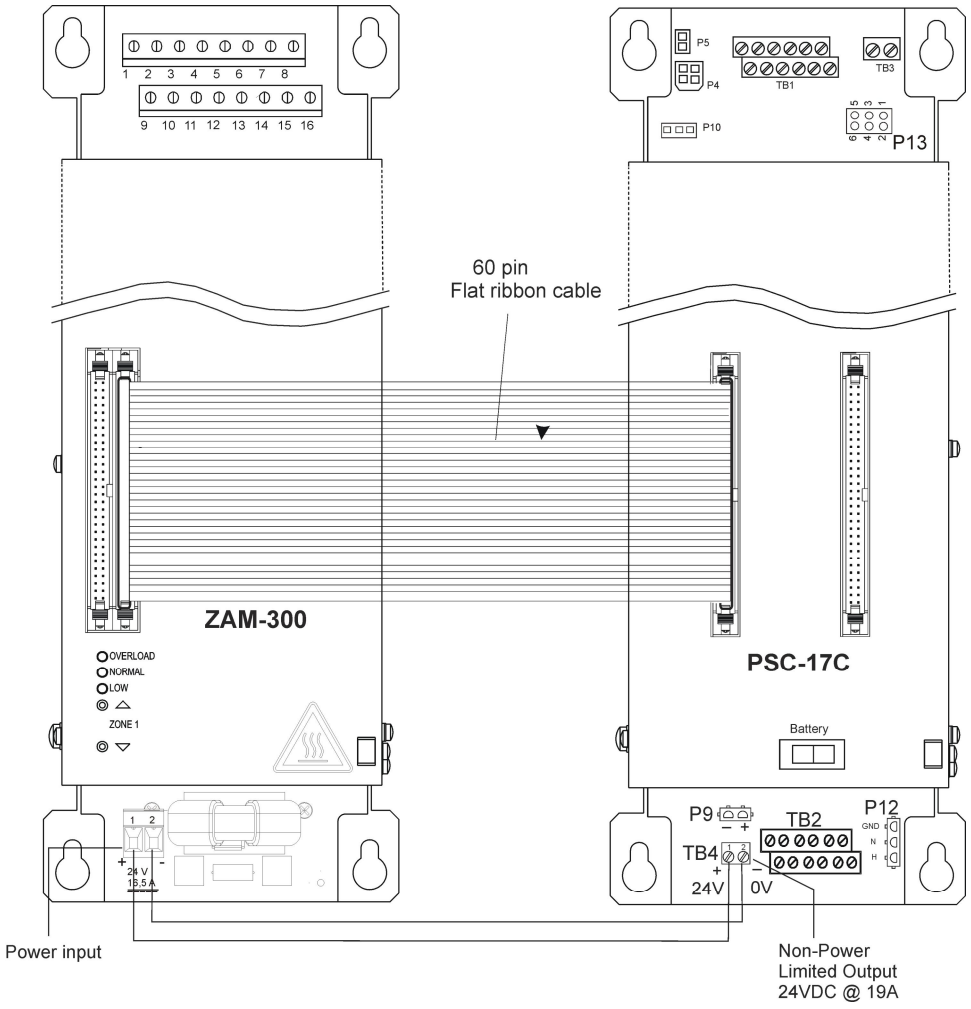
Lift the white cover on the terminal block.

Loosen the screw of the terminal by turning it counterclockwise.

Insert the wire into the side of the terminal block.

Tighten the screw of the terminal block by turning it clockwise

Power Input Wiring

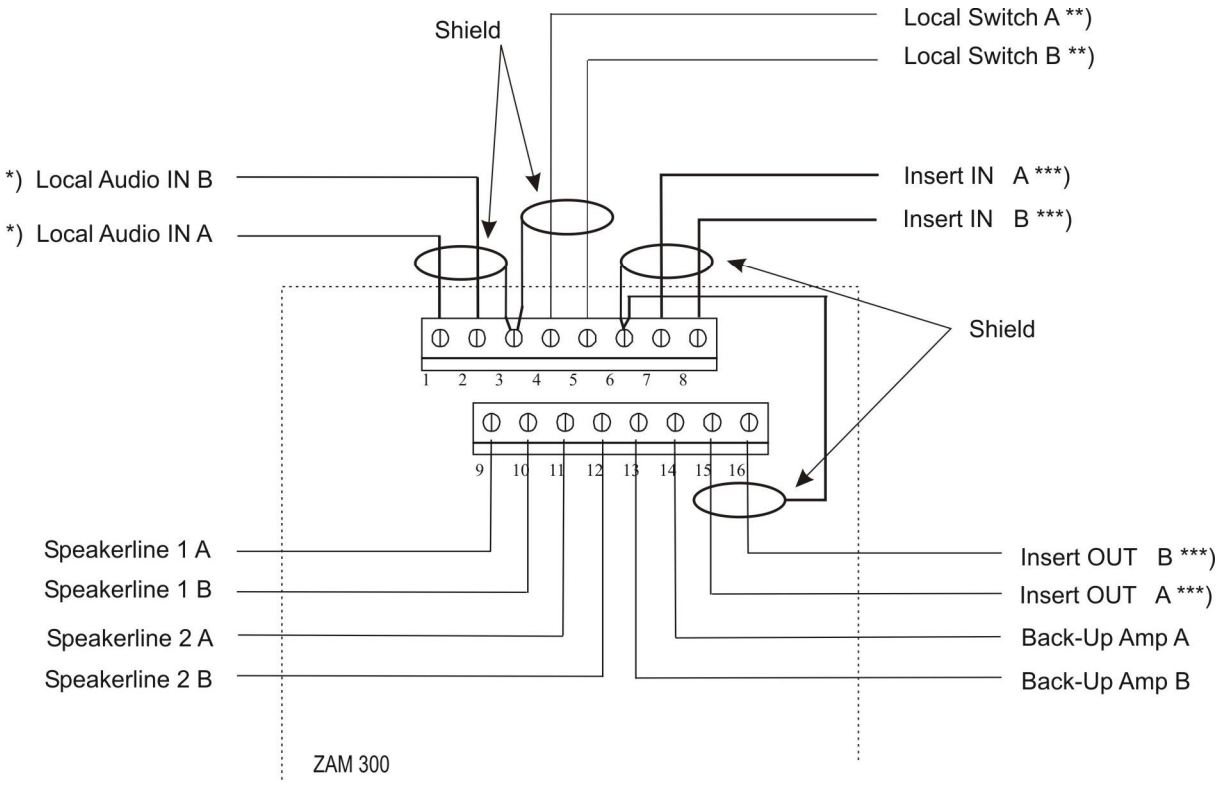


NOTES

1. 12AWG (2.5mm²) min.
2. Not power limited.

Figure 14 ZAM-300 24V Wiring

Local Audio Input, Audio Insert and Speaker Line Wiring



NOTES (Local Audio and Insert)
1. 24AWG (Ø 0.5 mm) min.
   12 AWG (2.5 mm²)max.
2. Use shielded twisted pair.
3. Terminate shields at one end only.

NOTES (Local Switch)
1. 24AWG (Ø 0.5mm) min., 12AWG (2.5mm²) max.
2. Use twisted pair or shielded twisted pair.
3. Terminate shields at one end only.
4. Connect to dry contacts only.

*) Balanced / synchronous input (isolated via input transformer)
Default setting: 0 dB = 0,775 VRMS

**) ZAM-300 will activate when input is shorted (dry contact).

***) Unbalanced (non isolated) 1,7dB
Input can be activated by setting jumper between pin 2 and 3 at socket board X109.

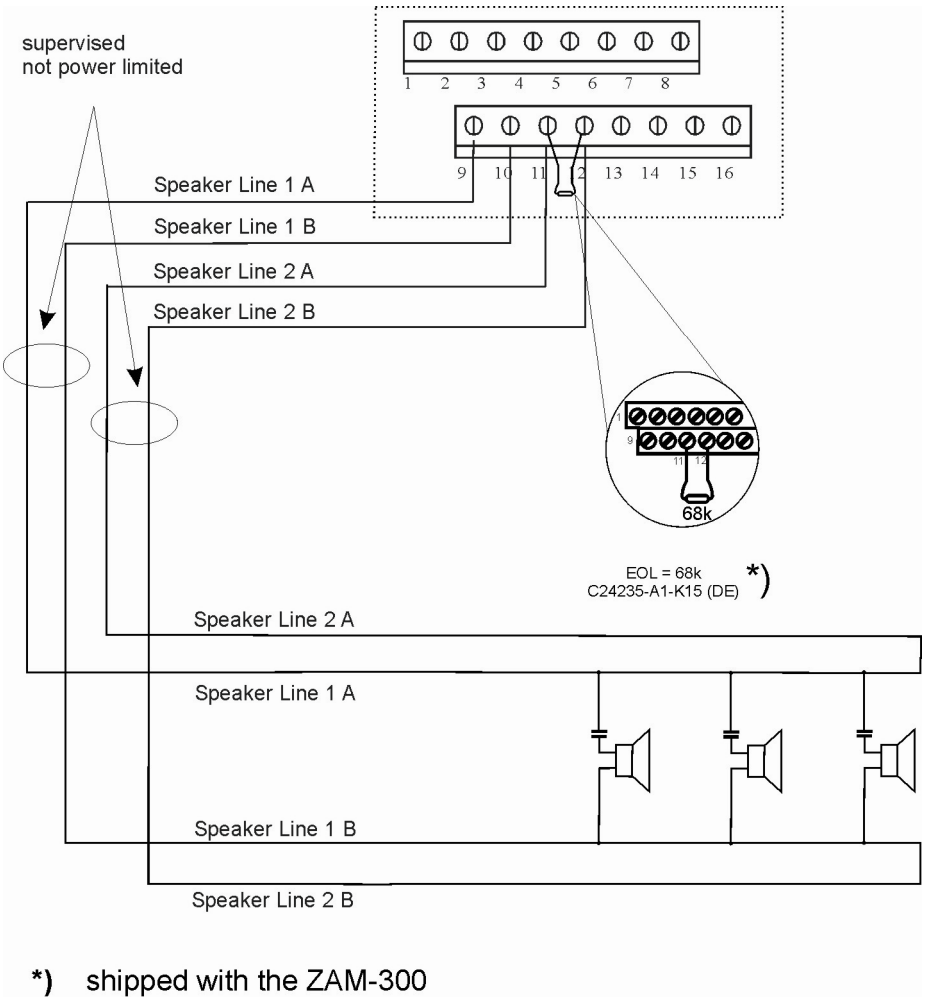
Figure 15 ZAM-300 Local Audio Input, Audio Insert and Speaker Line Wiring

Speaker Line Wiring

The configuration of the speaker lines can be configured to Class A (Loop) or Class A/B (two separate supervised and protected speaker lines).

Class A

The speaker line is looped back to the ZAM-300 terminal for the second output pair. The end of line resistor (EOL) 68 kΩ has to be connected at the end of the loop, directly at the ZAM-300 terminal.
As soon as an open is detected on the speaker loop the second output will be activated. Now the speaker line is powered from both ends.
A short circuit on the loop might overload the amplifier, dependant on the power of the audio signal and the resistance of the short.



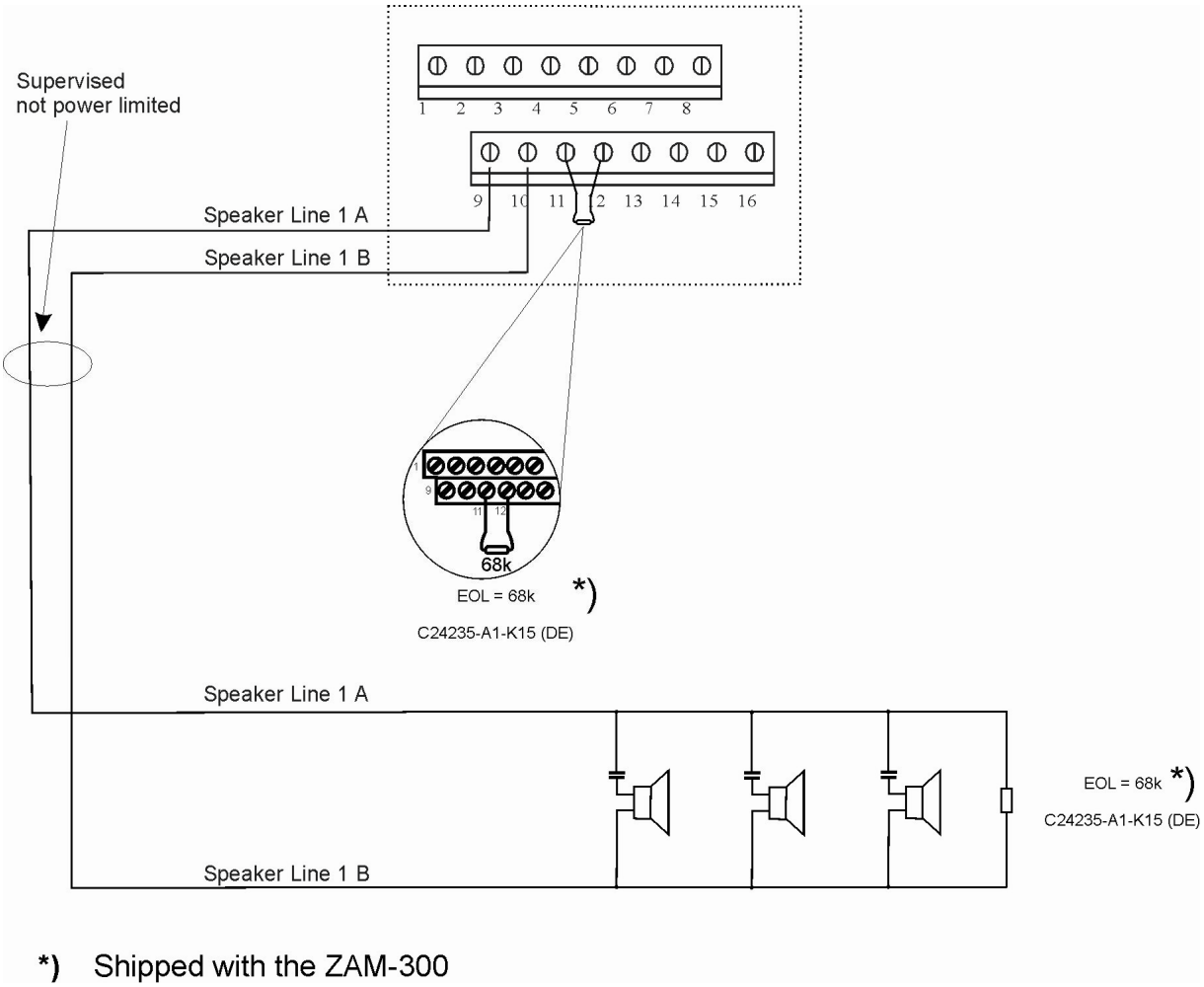
NOTES (Speaker Lines)

1. Use twisted pair or shielded twisted pair.
2. Terminate shields at one end only.
3. Supervised and not power limited.

Figure 16 ZAM-300 Class A (Loop) Speaker Line Wiring

Class B

Same as Class A/B, but second output of ZAM-300 is not used.
At the open output of the ZAM-300 the second end of line resistors (EOL) 68 kΩ has to be connected

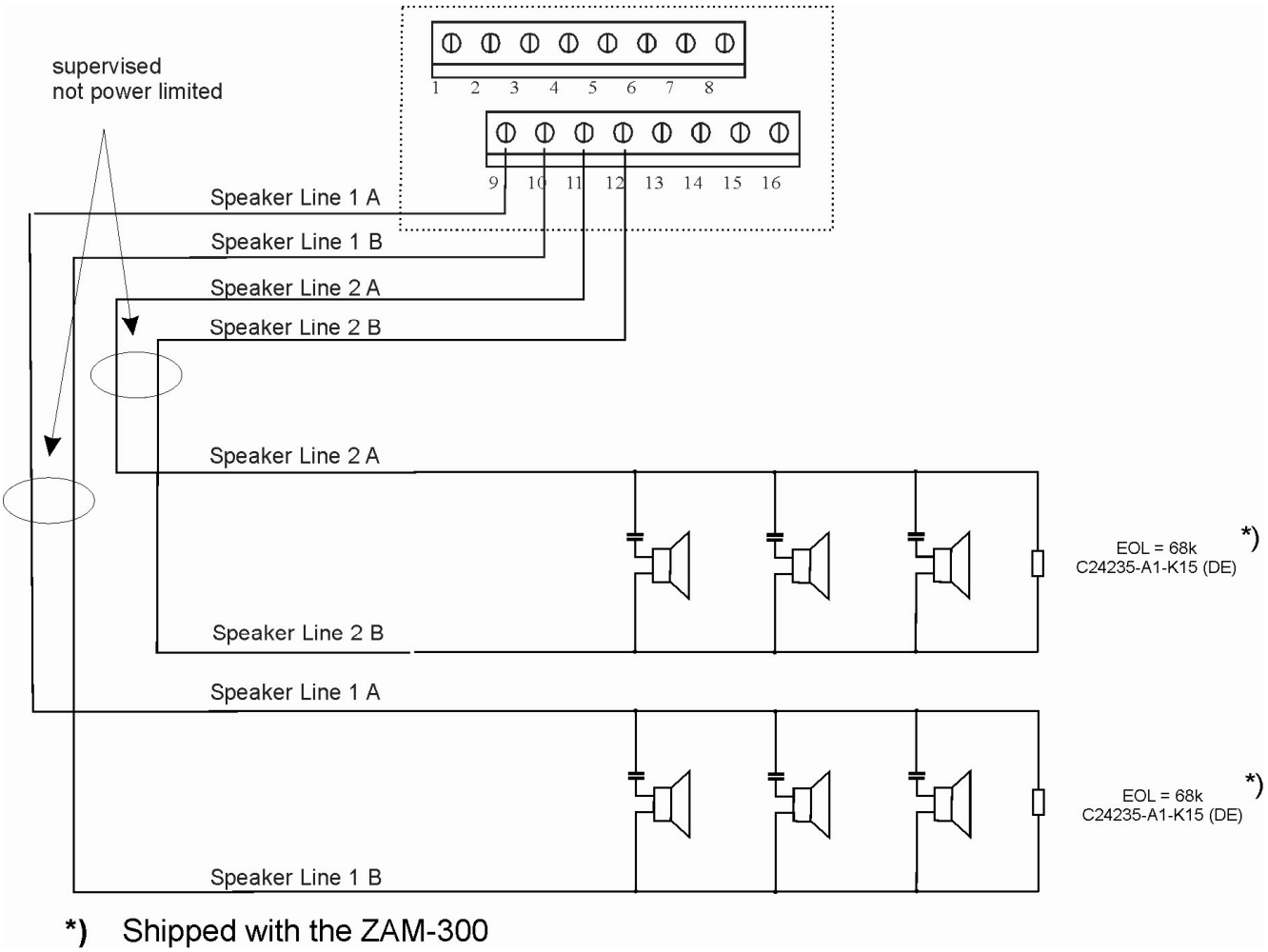


NOTES (Speaker Lines)
1. Use twisted pair or shielded twisted pair.
2. Terminate shields at one end only.
3. Supervised and not power limited.

Figure 17 ZAM-300 Class B (one Speaker Line) Speaker Line Wiring

Class A/B

Both speaker lines are used. The end of line resistors (EOL) 68 kΩ has to be connected at the end of each line - at the terminal of the last speaker.
From a users point of view the two speaker lines can't be activated/deactivated separately from each other. The two speaker lines build one speaker zone.
A hard short at one of the speaker lines might already overload the amplifier, dependant on the power of the audio signal and the resistance of the short.
A soft short at one of the lines with a resistance below approx 20 kΩ will cause the line supervision to detect a short.
Both types of short will result in a deactivation of this line or hinder this line from being activated. This functionality protects the second speaker line.
Benefit: only one ZAM-300 is enough to drive a Class A/B wiring installation
Typical application: speaker number 1, 3, 5, etc are powered by output one, speaker 2, 4, 6, etc are powered by output 2. If each room has at least one pair of speakers, the audio level will drop only by 3 dB in case of a short on one line.

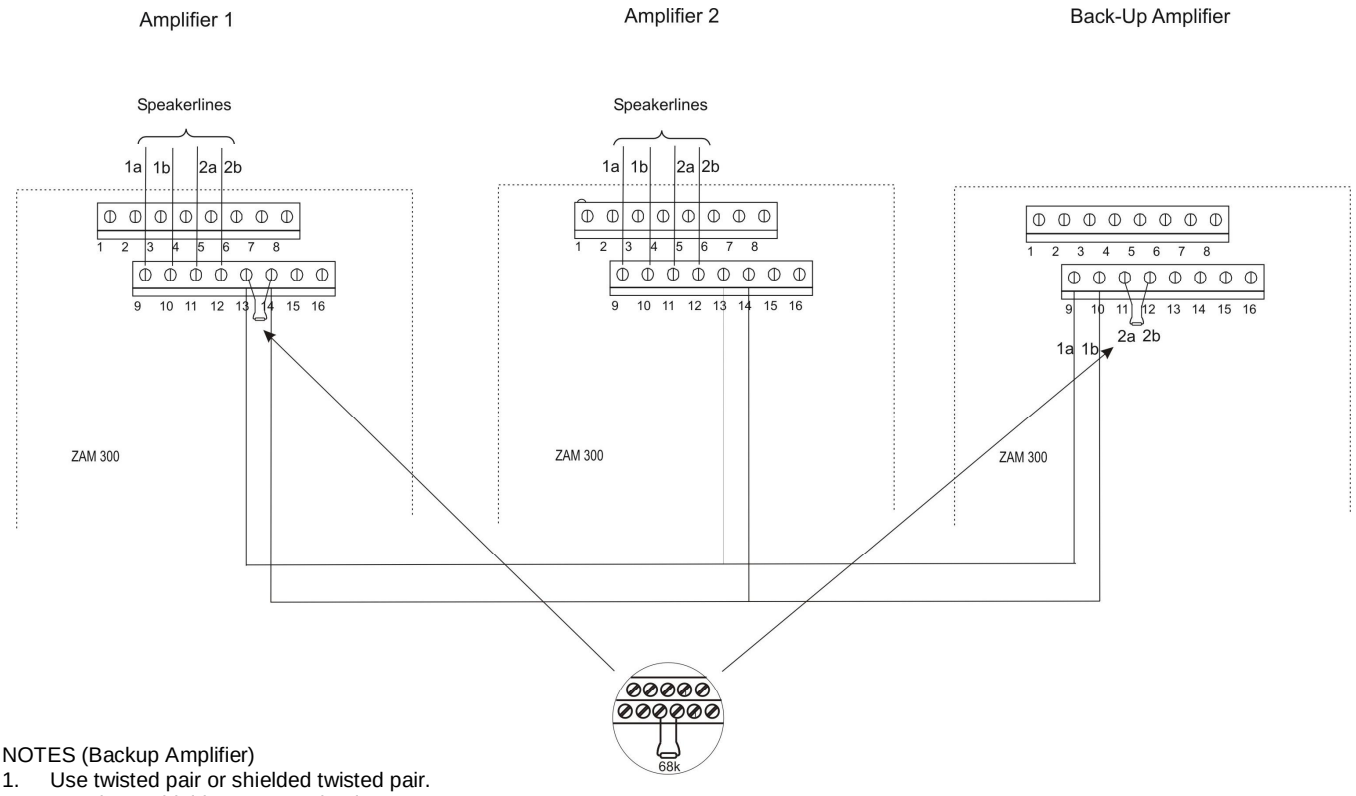


NOTES (Speaker Lines and Backup Amplifier)

1. Use twisted pair or shielded twisted pair.
2. Terminate shields at one end only.
3. Supervised and not power limited

Figure 18 ZAM-300 Class A/B (two Speaker Lines) Speaker Line Wiring

Back-Up Amplifier wiring



NOTES (Backup Amplifier)
1. Use twisted pair or shielded twisted pair.
2. Terminate shields at one end only.
3. Supervised and not power limited.

Figure 19 ZAM-300 Back-Up Amplifier wiring

In case of an ac main fault – running on battery – the amplifiers are automatically switched to the standby current reduction mode. This functionality helps to reduce the necessary capacity of the backup battery.

Input Power	
24V Back Plane Current	0 mA
6.2V Back Plane Current	0 mA
24V Standby Current	330mA
24V Standby Current while running on Battery	100 mA
24V Screw Terminal Current	Speaker Line Power x 0.045 + 24 V Standby Current = Total Current
24V Screw Terminal Current Reduction	Speaker Line Power x 0.045 + 24 V Standby Current Reduction = Total Current
300 W 300 W Reduction 200 W 200 W Reduction 100 W 100 W Reduction	300 x 0.045 + 0.3 = 13.8 A 300 x 0.045 + 0.1 = 13.6 A 200 x 0.045 + 0.3 = 9.3 A 200 x 0.045 + 0.1 = 9.1 A 100 x 0.045 + 0.3 = 4.8 A 100 x 0.045 + 0.3 = 4.6 A
Maximal Input Current at the 24 V input terminal	16.5A
Output voltage and current at the Dry Contact Switch Input terminal	24 V / 10 mA

For CE applications in E100 systems refer to
Installation Instruction A24205-A334-B844 (English) or A24205-A334-A844 (German)

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